

Geelong Beekeepers Club Inc.

*For hobbyist and
professional beekeepers*

General Beekeeping



In this collection

ARTICLES AND GUIDES

Bee Swarms	3
Bee Swarm Collection and Removal	4
Bees and Water	6
Biosecurity	7
Latest Alcohol Wash method	10
Drones 101	<i>Victorian Apiarists' Association</i> 11
Life cycle of honey bee	<i>Geelong Beekeepers Club</i> 12
Equipment Hire	13
Hive components	14
Hive components: approximate costs	16
Hive size calculator	17
Rearing Queens	<i>John Edmonds</i> 18
Register as a beekeeper	21
Requeening an aggressive hive	<i>Deb Smith</i> 22
Spring hive development	<i>Gary Montgomery</i> 24
7 ways to find the queen	29
Swarm control when you can't find the queen	<i>National Bee Unit (UK)</i> 32
Shook swarm	<i>National Bee Unit (UK)</i> 33
Total Fire Ban Days: use of smokers	38
What to do when you are Queenless	39
Winter pack down guide	42

VIDEOS

Flow Frame Cleaning	<i>Jurg Schutz</i> 45
Varroa Life Cycle	46
Winter welfare for your bees	47

What is honey bee biosecurity?	48
National Bee Biosecurity Program and Code of Practice	49
National Bee Pest Surveillance	50
Alcohol wash test for varroa mite	51
Sugar shake test for mites (Ag NSW)	52
Sugar shake test for mites (Ag Vic)	53
Drone uncapping test for varroa mite	54
Varroa: cultural control	55
Small hive beetle	56
Small hive beetle trap / attractant	57
Small hive beetle control devices	58

LONGER DOCUMENTS

Plan Bee: Honey Bee Breeding Manual	59
Apiary Code of Practice	60
Australian Beekeeping Guide	61
Australian Honey Bee Industry Biosecurity Code of Practice	62
Australian Honey Bee Industry Biosecurity Manual	63
Bait Hives for Honey Bees	<i>Thomas D. Seeley</i> 64
Bee Friendly: a planting guide for European honeybees and Australian native pollinators	<i>Mark Leech</i> 65
Fifty Years Among the Bees	<i>C. C. Miller</i> 66
Queen cells in my hive	67

ALSO

Varroa Management Hub – on the club website	›
Bee biology – in the club newsletter	›

Bee Swarms

What are Bee Swarms?

Bee Swarm - Teesdale (September 2014)

Bees swarming is a natural occurrence, which is most often seen in Spring. Bee swarms are identified by a mass of bees clustered on a post or hanging from a tree or some other structure. The bees may stay in this cluster up to a few days while they look for a new home.

Swarming bees are not dangerous - they are just looking for a new home and should be left alone. Do not spray them with a hose, smoke or insecticide as you may get stung. The bees will not harm you if they are left alone! Bees play an important role in fertilizing many of our nuts, fruit and vegetables.

Swarm removal



Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › Resources › General Beekeeping › Bee Swarms

Bee Swarm Collection and Removal

Please note - The Geelong Beekeepers Club is NOT a bee removal service. We MAY be able to assist depending on the circumstances, but as a group of volunteers, it's not possible for us to help in all instances.

Is it a swarm or an established/new hive?

Understanding the difference between a bee swarm and a hive or nest can determine the appropriate service you require for the bees to be removed safely.

Bee Swarms

Bee swarms are identified by a mass of bees clustered on a post or hanging from a tree or some other structure. The bees may stay in this cluster up to a few days while they look for a new home.

We recommend you contact one of the organisations or members listed on the club website as soon as possible, especially in relation to swarm bees, before they settle into a permanent place in your home - "Give them a call before they get into your wall".

Established Hives/Colonies

Bees moving in and out of a building wall, roof or chimney could indicate an established colony depending on how long they have been there.

If you can see honeycomb or the bees have been coming and going more than a week, chances are a hive is either well established or starting to establish.

For SWARM removal . .

Request assistance from a local beekeeper via the following sites.

- Victorian Apiarists' Association [Swarm Catcher Locator](#)
- International Swarm Patrol website - complete the online form. They have a number of beekeepers registered with the site who can contact you to arrange for removal. https://www.swarmpatrol.com/swarm_reports/add
- Aussie Apiarists Online - <https://aussieapiaristsonline.net/bee-removal.html>
- Bee Rescue: Geelong, Bellarine, Surfcoast & District Facebook group - <https://www.facebook.com/groups/809538203147190>

The Geelong Beekeepers Club is a volunteer organisation and has limited resources to assist with the removal of swarms and in particular established colonies. To assist the public with the safe removal or relocation of bee swarms and colonies, the above resources have been provided.

For Hive removal . .

To remove an established colony from inside a building (called a "Cut out") requires specialist skills and equipment as well as appropriate insurance. A suitably qualified Bee Removal Specialist will most likely charge a fee to remove the hive in this case.

Bee colonies can also exist out in the open where you will also see the honeycomb.

What can you do if you suspect an established hive in a building structure?

- Request help from a local beekeeper from the options listed previously.
- Engage a Bee Removal Specialist such as Bee Removals Victoria - <https://beeremovals.com.au/>

Club members who collect swarms

A list of Geelong Beekeepers Club members who may be able to collect free-hanging and accessible swarms in the Geelong district, with their areas and costs, is kept on the club website's [Swarm Removal page](#). Most do not do "cut outs".

For a beekeeper to determine their ability and capacity to assist you, it is helpful to provide the following information:

- Location (including suburb and where the hive is living such as inside a wall, chimney, roof or type of container such as compost bin)
- Provide a photo if safe to do so
- State how long the bees have been there
- Contact details such as phone or email

Depending on the difficulty or distance travelled to remove the bees, it is not unreasonable for the beekeeper to charge for their services. If it is an easy swarm removal, most beekeepers will provide a free service. We recommend asking the beekeeper if a fee will be charged before accepting their assistance.

Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › Resources › General Beekeeping › Swarm Removal

Bees and Water

Bees Need Access to Water

If honey bees do not have access to water, the beekeeper must provide it on the property on which the hives are managed.

This water must be placed, where possible, in a partially shaded position in close proximity to the hives. It is necessary to place this water in position before or as the hives are brought on to the property. This helps the bees to orientate to this water before they look for other sources elsewhere.

To allow the bees to alight and drink, water containers should include suitable floating material, or stones, bricks or sand to provide a beach effect if using a shallow pan. The use of Boardman feeders for water supply in urban areas is encouraged.

Source: Victorian Apiary Code of Practice, May 2011



Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › Resources › General Beekeeping › Bees and Water

Biosecurity

Early recognition of a honey bee pest or disease is one of the most important factors influencing the chance of controlling an outbreak and reducing its economic and social impact on the community.

There are several bee pests and diseases that have been declared as notifiable diseases under the [Livestock Disease Control Act 1994](#).

Diseases that must be reported without delay

- American foulbrood disease

Pests and diseases that must be reported within 12 hours

- Braula fly (*Braula coeca*)
- Varroa

If you suspect any of these diseases, you must notify an apiary officer: contact the Customer Service Centre on 136 186 or email honeybee.biosecurity@agriculture.vic.gov.au

Diseases that must be reported within 7 days

- Chalkbrood disease
- European foulbrood disease
- Nosema (*Nosema apis*)

If you suspect any of these diseases, you must notify an apiary officer: contact the Customer Service Centre on 136 186 or email honeybee.biosecurity@agriculture.vic.gov.au

Vic Agriculture [Online Public Reporting](#)

[Biosecurity Online Training \(BOLT\) Course login](#)

VISITORS

PLEASE RESPECT HONEY BEE BIOSECURITY

This apiary belongs to:

Beekeeper:

Contact:

Call in the event of an emergency
Do not enter this apiary without prior approval



Biosecurity recent topics at our monthly meetings

- [Wax Moth - John Bemelmans - September 2026](#)
- [Spring Opening - John Bemelmans - August 2026](#)
- [Nosema - John Bemelmans - July 2026](#)
- [Chalkbrood - John Bemelmans - June 2026](#)
- [IPM - John Bemelmans - May 2026](#)
- [Winter shutdown - John Bemelmans - April 2026](#)
- [AFB-EFB - John Bemelmans - March 2026](#)
- [Barrier system - John Bemelmans - February 2026](#)
- [Small Hive Beetle - John Bemelmans - November 2025](#)
- [Wax Moth - John Bemelmans - September 2025](#)
- [Spring Opening - John Bemelmans - August 2025](#)
- [Nosema - John Bemelmans - July 2025](#)
- [Chalkbrood - Feeding - John Bemelmans - June 2025](#)
- [Winter shutdown - John Bemelmans - April 2025](#)
- [Preventing Disease - John Bemelmans - November 2024](#)
- [Small Hive Beetle - John Bemelmans - September 2024](#)
- [Spring - John Bemelmans - August 2024](#)
- [AFB-EFB - John Bemelmans - July 2024](#)

- [Nosema - John Bemelmans - May 2024](#)
- [Wax Moth - John Bemelmans - April 2024](#)
- [Integrated Pest Management - John Bemelmans - February 2024](#)
- [Nosema - John Bemelmans - May 2023](#)
- [Apiary Hygiene - John Bemelmans - April 2023](#)
- [Braula fly and Varroa - John Bemelmans - October 2022](#)

Other National/state Biosecurity news stories

Biosecurity factsheets

Biosecurity and Bee Pest Surveillance

Biosecurity Videos

Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › [Resources](#) › [General Beekeeping](#) › [Biosecurity](#)

Latest Alcohol Wash method

Note: We now know that the sugar shake results in significant mortality and is less reliable than alcohol (or soapy water) washing.

The combination of shaking and the sugar blocking the bees' respiratory spiracles causes death (not necessarily immediately), remembering that shaking is required to dislodge the mites from the bees to allow detection.

Another non-lethal method of mite monitoring is to measure mite drop using a sticky mat and this is reportedly popular in the UK. Interpretation of the results is much trickier, varying substantially with size of the colony and time of year. Because of this, the National Varroa Mite Management Program has not recommended its use. Instead, alcohol wash is the recommended method.

Allerton, M., 2026. Sugar Shake - Let's Stop Sugar Coating It: The Powdered Sugar Shake Is Not Harmless. Australian Bee Journal, 107 (1), 8-9.

Bruckner, S., Williams, G.R., Tsuruda, J. and Underwood, R.M., 2025. Let's not sugar coat it: the powdered sugar shake is not harmless for honey bee workers. Journal of Apicultural Research, pp.1-6.

Alcohol wash for varroa — Tocal College / NSW DPI



The video plays here in the reader. On paper: [geelongbeekeepersclub.org.au/edit/movie/alcohol wash - tocal nsw dpi.mp4](https://geelongbeekeepersclub.org.au/edit/movie/alcohol%20wash%20-%20tocal%20nsw%20dpi.mp4)

Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › Resources › General Beekeeping › Latest Alcohol Wash method

Drones 101

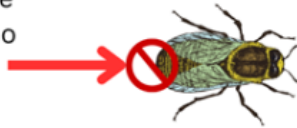


Drones 101

Honey bee drones: chubby freeloaders or crucial hive citizens? Let's take a closer look at the male honey bee.



Drones have no stinger so they can't defend themselves.



Drones have double the mass of a worker bee but about the same as a queen. They are longer than a worker bee and slightly shorter than the queen.



Workers weigh about **0.11 grams** and are about **15mm** long.

Queens weigh about **0.18 - 0.2 grams** and are **20-25mm** long.

Drones weigh about **0.2 grams** and are about **20 mm** long.



Drones typically do not mate with a virgin queen from the same colony to ensure genetic diversity. Inbreeding has been shown to result in smaller, shorter-lived offspring that show less pollen and nectar hoarding behaviour and neglectful brood-rearing tendencies. Worker bees have been known to eat in-bred brood. This prevents them from hatching and recycles the protein in the hive.



For the first few days of their lives, drones are fed by worker bees. After they are about a week old, they can feed themselves from honey stores.

A drone can fly for about 20 minutes on a mating flight before having to return to the hive to refuel on honey. It takes a drone about 15 minutes to fill up on honey before doing another flight.



A drone's eyes are more than twice the size of those of a worker bee. The large eyes allow the drone to spot a virgin queen flying through a mating congregation zone.



Drone

Worker



Drones mate with a queen in mid-air. The force of the mating causes the drone to instantly pass out. The drone then falls to the ground and dies.

Drones that do not mate stay in the hive. It is thought that drones, due to their large body mass, assist with hive thermoregulation while in the hive. In late autumn, however, they are starved and ejected from the hive by worker bees. It is estimated that fewer than 5% of drones actually mate with a queen.

Life cycle of honey bee

Life cycle of honey bee

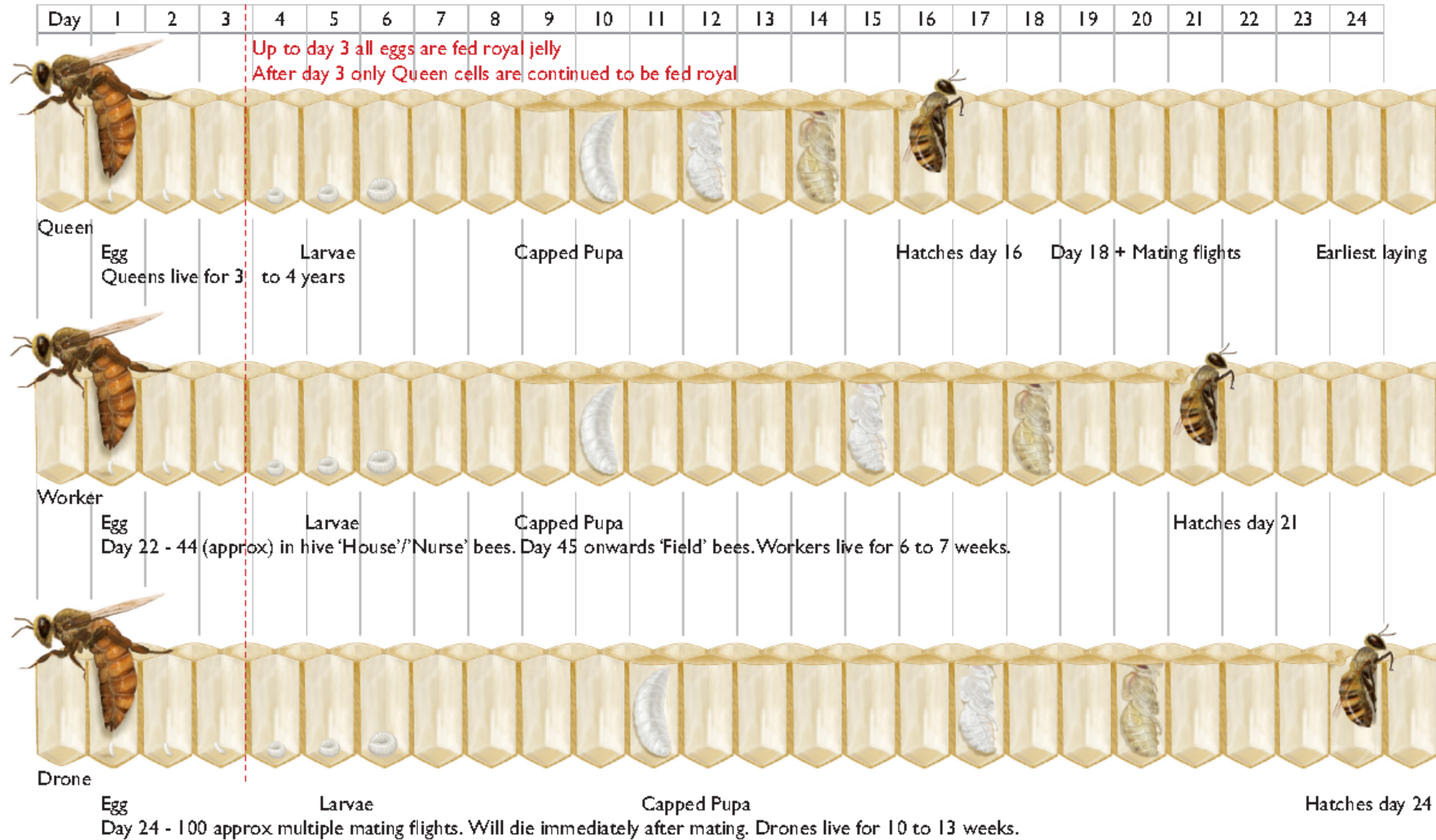


Chart: Geelong Beekeepers Club · from the club website

Equipment Hire

Extractor Hire

The Geelong Beekeepers Club has two extractors available for hire to members only.

Extractor hire fees:

3 frame extractor \$20.00 per FIVE (5) days + \$50.00 deposit

2 frame extractor \$10.00 per FIVE (5) days + \$50.00 deposit.

One 3 frame extractor and one 2 frame extractor are located at freshwater Creek.

One lighter 2 frame extractor is located at Batesford.

Wire jigs

Wire jigs hire fees: \$5 per week + \$20 deposit (Located Batesford).

It is a condition of use that this equipment is returned in a clean, wax and honey free condition.

The Geelong Beekeepers Club accepts no responsibility or liability for the transmission of any disease acquired by the use of this equipment.

Contact Stephen Fridey to arrange hire and collection - sfridey@icloud.com or use the form below to request an extractor. We will contact you to confirm the booking and collection arrangements, or to arrange an alternative date.

Request equipment hire

[Online request form - on the club website]

Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › Resources › General Beekeeping › Equipment Hire

Hive components

Typical LANGSTROTH hive

The hive body or boxes holds 8 to 10 frames that are standardized in length. The frames hold the foundation and the honeycomb that is built on it.

Hive equipment manufacturers will often produce bodies and supers that vary slightly. The differences between manufacturers are generally 2mm or less.

A 'Queen Excluder' is a screen or mesh of a specific size that allows normal bees to pass through, but because the Queen is slightly larger, she can not pass through.

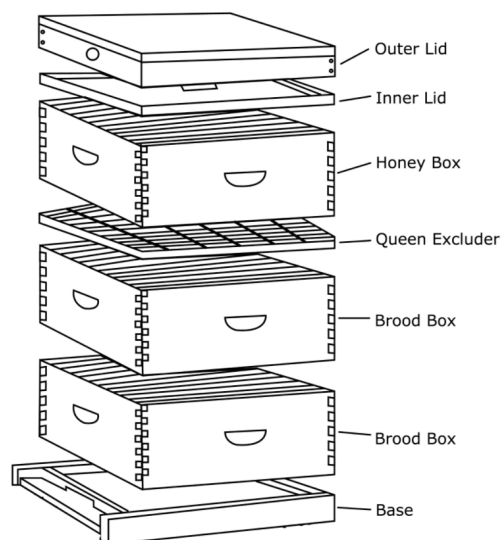
Placing a Queen Excluder between a brood box and a honey box means that while the brood box may contain both brood and honey, the honey box above the Queen excluder will only contain honey as the Queen can not get through to lay eggs.

A 'Brood Box' and a 'Honey Box' from a construction point of view are the same physical box. The difference is just in the positioning and function within the hive.

The 'Brood Box(s)' at the bottom of the hive is where the Queen lays her eggs and the young bees (brood) are raised.

'Honey Box(s)' (sometimes called 'Honey supers') at the top of the hive are where honey is stored. Honey is also stored in the lower Brood box, and if a Queen Excluder isn't used, some brood may be placed in the Honey boxes.

The following table includes approx interior dimensions.



Type	Depth	Interior length	Interior Width
10 Frame Deep body	240 mm	470 mm	370 mm
10 Frame Medium body/super	170 mm	470 mm	370 mm
10 Frame Shallow super / Ideal	144 mm	470 mm	370 mm
8 Frame Deep body	240 mm	470 mm	310 mm
8 Frame Medium body/super	170 mm	470 mm	310 mm
8 Frame Shallow super / Ideal	144 mm	470 mm	310 mm

FLOW Hive

Flow hives are essentially a 'honey extraction system' and are very similar to Langstroth hives as far as base and brood box is concerned. The big difference is in the honey boxes/super's and frames which are a unique design that allows the honey to be extracted by turning a tap and without the need of removing the frame from the hive.

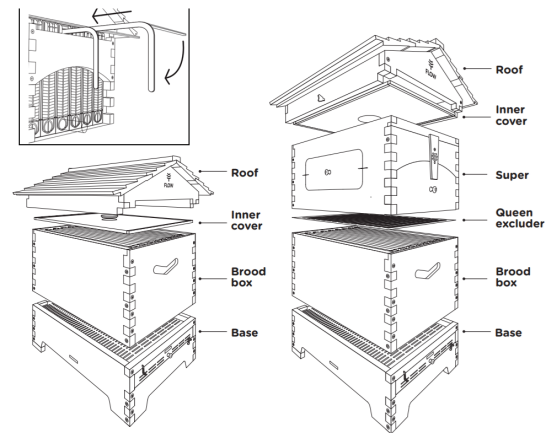
To extract honey from traditional (Langstroth) hives, the frames are removed from the hive and the wax covering the honeycomb is cut/scraped off and then the honey is spun or dripped out.

To extract the honey from a Flow hive, a lever is inserted into the honey frame to temporarily shift the position of the plastic comb components allowing the honey to flow out a tube inserted at the bottom.

The advantage of this system is that the frames remain in the hive and there is minimum disruption to the bees. There are however a few disadvantages to this approach as well. It can be difficult to see if the comb is capped and the majority of the honey ready to extract. If the honey is thick or crystallised it will not flow out, and the frames will need to be removed from the hive and heated to enable the honey to flow.

Flow hives are considerably more expensive than traditional Langstroth hives.

Approximate costs



Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › Resources › General Beekeeping › Hive components

Hive components: approximate costs

(updated 3/2026 - For up to date Flow Hive pricing go to <https://www.honeyflow.com.au/collections/all>)

Component	8 Frame Langstroth	10 Frame Langstroth	Flow Hive
Top/Lid	\$30.00	\$32.00	
Ideal box	\$26.00	\$24.00	
Ideal frames	\$1.70 ea - \$13.60	\$1.70 ea - \$17.00	
Queen excluder	\$18.00	\$20.00	\$1370 (complete hive)
Full height box	\$27.50	\$32.00	
Full height frames	\$2.00 ea - \$16.00	\$2.00 ea - \$20.00	
Vented base	\$45.00	\$48.00	
Smoker	\$65	\$65	\$65
Hive tool	\$10	\$10	\$10
Suit/Veil	\$180	\$180	\$180
Approx total	\$435.00	\$450.00	\$1625.00

Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › Resources › General Beekeeping › Hive components approximate costs

Hive size calculator

A couple of different methods. Disclaimer - neither are particularly accurate! and at best only an approximate guess.

Method 1 - Observe the bees seen leaving the hive

- On a reasonable foraging day, only two thirds of the bees are inside the hive.
- Count the number of bees leaving the hive in one minute.
- Arizona University researchers came up with this formula: $N = (f / 0.0138) \times 3$ (where N = the approximate number of bees in the hive)
- f - foraging rate: count how many bees exit the hive in a single minute during peak foraging hours (typically mid-morning or early afternoon on a warm, clear day).
- The constant (0.0138) represents a mathematical factor based on typical foraging trip durations and flight frequencies per individual bee on an average day.
- The multiplier (3) accounts for the general rule of thumb that actively foraging field bees make up roughly one-third of the total adult population in a healthy, established colony.

The formula can be mathematically simplified to $N = f \times 217.4$. For example, if you observe 56 bees leaving the hive in a minute, the approximate number of bees in the hive is $56 \times 217.4 = 12,174$.

Method 2 - Visual estimating

Enter the number of frames in your hive that match the reference photos on the calculator page (frames of roughly 500, 600, 700, 800, 900 and 1,000 bees per side) and the calculator adds them up. For example 3 frames of 500, 5 of 600, 5 of 700, 6 of 800 and 1 of 900 gives an estimate of 41,100 bees. You can print the photos so you have a visual reference while inspecting your hive.

The interactive calculator lives on the club website.

Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › Resources › General Beekeeping › Hive size calculator

Rearing Queens

By John Edmonds

A new queen each year will result in strong healthy bee hives, and if of docile manner it makes working the bee hive a much quicker and more pleasureable pastime.

Most important is to have a quality breeder queen. Undesirable traits can be magnified if an ordinary queen is used. In nature bees tend to revert to aggressive traits as they are more successful in protecting the colony.

Mature drones of the correct age must be present before making the queen cells. When drones are hatching then queen cells can be started. Drones require 14 days and have started flying to drone congregation areas.

Good floral conditions and supplementary feeding will result in premium queens. It is best when the bees are gathering a steady flow of mixed sources pollens and a steady flow of nectar. It is important that the bees can source pollen from plants of European sources, although I now think that supplementary feeding Agrisea also provides the required ammino acids. Heavy honey flows will often see the bees refuse to make queen cells. Sometimes bees are not inclined to be in the mood for making queen cells and will chew down the started queen cells if the honey flow stops, weather is unsuitable and possibly cold nights.

Making a queenless hive will cause the bees to start drawing queen cells. I recommend after 2 or 3 days you knock down the queen cells they have started from their own brood and collect the royal jelly with an artist paint brush.

Wash the plastic queen cells to remove any oil film from the molding process. Best to place the queen cells into the hive for a day prior to grafting for the bees to prepare the queen cells and warm them. Using the artists brush smear a layer of royal jelly into the base of the queen cell cups.

Grafting the grubs require selecting the 12-to-24-hour grubs. (a small "c") It is important not to use the big grubs as although the bees will make queen cells the queens will hatch earlier and kill her sisters and be of poor quality. When the bees draw out queen cells from older grubs, the older the grub, the more worker bee features the queen will have and the less ovaries.

Using a Chinese grafting tool lift the grub out of the frame cell and transfer to the plastic queen cell on top of the royal jelly. Black plastic frames are much easier to graft from, and newly drawn frames are the hardest. If the grubs are being fed profusely and resting on top of plenty of bee milk then it is much easier to graft the grubs. Feeding sugar syrup, the night prior will enhance the production of bee milk.

I prefer to graft late afternoon and take care that the strong sunlight does not burn or dry out the grubs. I graft the cells at the apiary at the back of the Ute so it is important to have quiet bees as it is easier if you do not have to wear a veil. If the day is hot, I return the frame of selected brood back to the breeder hive after a short time and select another frame so the brood is fresh and not over heated. Usually, the sunlight is ok for good vision but during cloudy days or if you are working into the evening the wearing of a forehead light makes it easier to see the grubs.

I prefer to draw approximately 16 queen cells per 8 frame full depth box of bees. I often see were beekeepers draw many more queen cells but they will be inferior size small queens.

I like to feed the bees approx 1 to 1.5 litres each day of 50/50 sugar syrup with Agrisea for the first few days, this will result in more queen cells and better-quality queens. A leading Beekeeping researcher said "do not buy queens from a queen breeder that does not feed the bees".

It is most important to keep good records of when you have made the graft as in 10 days from grafting the queen cells will be ready to place into the nucleus hives you have made up. I find in strong warm hives the cells will hatch approximately midday on the 11th day. In smaller less populous hives, the queen cells may take an extra day to hatch.

When removing the queen cells from the bee hive it is important to be gentle and do not shake the bees off the frame or jolt the queen cells during transport. The queen's wings are the last thing to develop and are sensitive to rough handling.

I have been told do not place the queen cells on their side, must be kept upright but I have found them to be still OK. Queen cells that are about to hatch are remarkably tough and can stand cooling slightly and this will slow down the hatching time.

It is important to position and colour the nucleus bee hives so the queens can recognize and remember the position and not stray into a foreign bee hive. Returning virgin queens have a bad habit of drifting into your breeder hives and killing your breeder.

Nucleus bee hives are generally best if they are 1 frame of honey and stored pollen, 2 frames with brood, and an empty frame making a 4 frame nuc.

Press the queen cell into the beeswax of the frame just above the brood and the bees will keep it warm. It can be a good idea to use a conical cage to protect the queen cell or even the use of electrician's tape around the side of the queen cells will prevent worker nasty bees from rejecting the new queen. The emerging queen communicates with the worker bees and sometimes must say the wrong thing and is killed. Rarely the queen cannot hatch as she is upside down in the queen cell.

The queen will emerge and takes about 7 to 10 days to mature for her mating flights. She will wait for days above 18/20 degrees and little wind. She will fly further than the drones from her colony to drone congregation areas that remain in use every year. How and why these drone congregation areas exist is a mystery. 20,000 drones will circle around these drone congregation areas waiting for the virgin queen to fly through the drone congregation area and the fastest and fittest drones will mate in the air approximately 60 meters above ground. It only takes 2 seconds for the mating to take place and the drone dies instantly falling backwards dead. The next drone removes the remnants of the previous drone and research shows approximately 15 drones will mate over one to two days with the queen. Usually, the queen only absorbs 10% of the semen. This will last her for her lifetime. Drones fly a short distance, whereas the queen flies a longer distance to mate. Drones can only stay away from the hive for 25 minutes and have to return for more fuel (honey) and fly 4 to 6 times per day.

After 21 days it is a good time to inspect the nucleus and see if the queen has succeeded. Approximately 70% of the queens will be successful.

I prefer to wait 28 days plus until the brood starts hatching so I can see the results of the queens mating. Research has shown that queens caught from the 28-day nucs have a better success rate in being accepted in the new hive.

Another method for making queen cells is to have a double decker hive with only 4 frames of honey, pollen and sealed brood, the rest of the box has 4 follower boards (timber cut into the shape of frames that reduce the space in the box). The bottom board has a mesh base to allow the bees air. The entrance is closed. The young bees from a strong double hive are shaken into the bottom box the older flying bees will return home to the queen. The bees will cover the 4 frames and the rest hang as a cluster like a swarm under the frames into the empty chamber below. The graft of queen cells is introduced and the bees kept locked in for 2 days in a cool place (often inside a shed).

The bees panic and start making queen cells. The queen cells can then be transferred to a strong bee hive that has a super over a queen excluder with honey and pollen frames and the center has unsealed brood to attract the nurse bees through the queen excluder. The bees will finish the queen cells. It is thought that this method makes the best queens.

Cloake method uses a strong double decker hive that has a queen excluder and cloake board which has a slide and upstairs entrance. The slide can be inserted making the upper super queen less, the bees will start drawing the queen cells and then after 2 days the slide is removed to allow the hive become a complete unit and finish making the queen cells.

Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › Resources › General Beekeeping › Rearing Queens

Register as a beekeeper

Anyone who owns one or more hives of bees in Victoria is required by law to register as a beekeeper with the Department of Agriculture.

Registration is free for up to 5 hives.

Go to <https://bees.agriculture.vic.gov.au/Home/Registration>



Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › Resources › General Beekeeping › Register as a beekeeper

Requeening an aggressive hive

By Deb Smith

Depopulating the Brood Box Method

EQUIPMENT NEEDED

- Spare base and lid
- Spare box (full depth box if your brood box is full depth)
- New queen (preferably mated in a cage from a reputable breeder)

Notes before you start

In this scenario, the hive is made up of a full depth brood box, queen excluder and honey super. Therefore, you know before you start, the queen is in the bottom brood box. If you have more than one honey super on the hive, ensure all honey supers are left in the original hive location and you are only moving the brood box away from the area to find the queen. Requeening is more efficient with two sets of eyes. Engage the help of a friend if you can!

METHOD

1. Smoke the hive as usual.
2. Depending on the space around your hive, either move the whole hive back a little and rotate 180 degrees, or if moving the brood box 20 metres away at this point, move the whole hive to the side.
3. Place the spare base board in the original hive location.
4. Place the honey super on the spare base board and add the spare lid.
5. Place the hive lid on the brood box.
6. Depending on your decision to either rotate the hive 180 degrees or move the brood box 20 metres or so in the above step, follow one of the applicable steps below:
 - (a) If you have turned the brood box 180 degrees, leave the box for a couple of hours so the foraging bees fly back to the original hive. This will depopulate the brood box and you will have far less bees to deal with to find the queen. The foraging bees tend to have the attitude too. After a couple of hours, move the brood box away to find the queen.
 - (b) If you are moving the brood box away from the original location, use a wheelbarrow or trolley to assist with easy movement, then leave the brood box for a couple of hours.
7. After a couple of hours, it's time to find the old queen. There are several ways to do this. My preference is to start searching on the outer frames and work my way towards the centre. Have your spare box set up next to the brood box on top of the hive lid so you can place checked frames in this box. With the sun over your shoulder, start with the closest frame and check the dark side first which is the opposite side to what you see when you lift the frame. Scan your eye beginning in the top left corner and move across the frame, then continue back across to the left until you're satisfied, she's not there. Turn the frame over and check the other side. Place the checked frame in the spare box, same location. Continue checking frames in the following number order until you find the queen: 8, 1, 7, 2, 6, 3, 5, 4.

8. When you find the queen, despatch her, then return all frames to one box, add the lid and move the brood box back to the original location to reassemble the hive and add the new queen.
9. At the original location, remove the lid, use some smoke as these are the feisty foraging bees and place the honey super on the up turned lid.
10. Return the brood box to the original location and add the new queen. Remember to cut the appropriate end of the cage so the candy plug is exposed and facing upwards. Place between frames towards the middle and ensure it won't slip.
11. Add your queen excluder, honey super/s and lid.
12. Monitor for acceptance. Wait at least 5-7 days before checking the new queen has been released from the cage and is laying eggs.

What happens if you did not find the queen?

One method is to spread a sheet in front of the hive up to the hive entrance before reuniting the brood box and super. Shake all bees from the brood box onto the sheet.

Place a queen excluder on top of the hive base, then add the brood box with brood frames. The bees will walk back into the hive to look after the brood. Scan for the queen on the sheet, but if you still can't find her, she will follow the bees back into the hive. The workers will be able to move up through the queen excluder back to the frames, but the queen (and drones) will be trapped between the base and queen excluder. Leave the bees for an hour or so or until they have all returned to the box before removing the brood box and queen excluder to check the base for the queen. Hopefully now she will be easy to see! Despatch her, add the new queen and reassemble the hive.

Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › Resources › General Beekeeping › [Requeening an aggressive hive](#)

Spring hive development

By Gary Montgomery (Photos credits: Aust Beekeeping Guide; Vic Gove Ag Notes; Cashmanbees.com)

If you have followed the appropriate winter hive management practices covered in these notes over the past two months including checking the hives intermittently to assess the need for any supplementary feeding by "lifting" them from the rear, your hives should have "wintered" successfully.

Now, proper hive management in early spring (September) is critical for maximizing success for the forthcoming season.

Your hives now should be very carefully assessed in relation to the factors listed below.

First inspection after winter

This should only be carried out when it is a relatively warm day (18 degrees +), there is little wind and worker bees are flying. Opening the hive when it is too cold will allow warm air to escape, cause the winter bee cluster to "break-up" as you remove frames for inspection and causes stress to the bees.

All of these factors may result in a setback to brood development and increased rate of use of the remaining honey stores in the hive, as the bees will need to consume more food to reheat the hive.

When you do open the hive for inspection, carefully adjust the hive mat on the top of the frames, to ensure it covers those frames you are not inspecting, to retain as much warmth as possible in the hive.

Stores of honey and pollen

At the beginning of springtime, honey stores in the hive are at their lowest level. Foraging bees from the hive will have become increasingly active, consuming more of this food store for energy.

With increasing amounts of pollen and nectar being brought into the hive, the queen will be increasingly stimulated to lay, resulting in significant brood development. For one comb of brood to successfully develop, one full comb of honey, plus pollen stores, is necessary.



As this brood develops and emerges from the cells, the newly emerged young bees require some of the increasingly scarce stores of honey, although are not yet able to forage and contribute to its replenishment. This puts significant pressure on the remaining scarce food supply inside the hive.

The sudden growth of hive numbers, along with diminishing supplies of food, may result in what appeared to have been a strongly developing hive, suddenly being significantly depleted in numbers, as bees die from lack of food. This is referred to as "spring dwindle." Careful checking of the available food stores at the start of spring can avoid this problem.

Supplementary feeding by use of sugar syrup (see June “In the Hive”); shifting the hive to a location where adequate nectar flow is available, or taking a “spare” frame of honey from one hive and transferring it to one that is in need of food, are all viable options. If the latter process is used, extreme care must be taken to ensure you are not transferring disease from one hive to another!

The Queen and Brood

It is essential to check as early as possible in spring that the hive has a fertile and vigorous queen. This can be assessed by observing the laying pattern established. If there is a uniform pattern of eggs, larvae and sealed brood, the queen is performing up to expectation and should be retained. There is no need to actually sight the queen.



If eggs appear to be on the side, rather than in the middle at the bottom of a cell, this indicates an old / failing queen. Similarly, if there are an abnormal number of drone cells evident (raised bullet shaped capping) the queen is losing her fertility and the hive should be united with another. It is not often that there will be sufficient brood to enable the colony to survive if the queen is a drone layer, even if a new young queen is introduced.

If the brood laying is scattered (as at right), this may be an indication of an aging / failing queen, or the presence of disease.

Presence of diseases

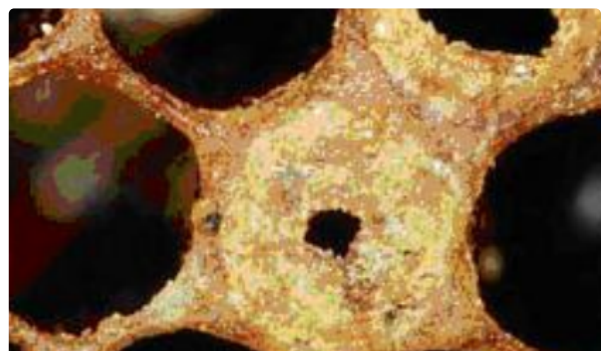
It is vital to check for any signs of disease early in spring. A very slow or negligible build up in brood numbers at this time may be an indicator of the presence of AFB or Nosema. Both of these may result in the complete loss of a hive.



a) AFB. - A virulent, destructive and notifiable disease. Although difficult to detect in the early stages, detection off this disease is important to prevent its spread to other hives which will also succumb to the disease! It may be found in both weak and strong hives!!

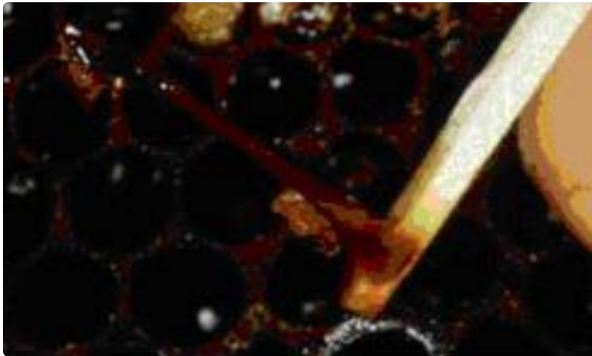
Symptoms of AFB in the Brood.

- AFB diseased larvae and pupae always lie stretched out on their backs on the lower wall of their cells, from the back of the cell (the mid-rib of the comb) to the cell opening. If the cell is likened to a clock-face, the diseased larvae / pupae will be positioned at the bottom of the face between the figures 5 and 7.
- The caps of cells containing diseased larvae or pupae may be sunken, concave, dark and at times greasy-



looking. Some caps may be perforated and other caps may be totally removed.

The match stick test method of diagnosing AFB is very effective - To do this insert a matchstick into the moist, decaying remains of a dead larva may be which may then be slowly withdrawn from the cell. The remains will stretch approx. 25 mm or more.



The Livestock Disease Control Act 1994 requires that anyone who knows or suspects that AFB is present in his or her apiary, must notify a DEPI apiary officer within 12 hours.

b) - Nosema. - Bee diarrhoea. Evidence of this is seen in the spotting of excreta on combs and inside hive walls. This can lead to further spread of the infection. Nosema may significantly weaken bees which may be seen crawling on the ground in front of the hive, and attempting to

fly. Usually they die some distance away from the hive. Although this problem can significantly deplete a hive, improvement in the weather may rapidly assist a colony to recover.

Nosema may also contribute to "spring dwindle". - An "unexplained" loss of numbers in the hive, or an apparent failure of a hive to develop as rapidly in springtime when one may expect otherwise!

Source: <http://agriculture.vic.gov.au/agriculture/pests-diseases-and-weeds/animaldiseases/bees/nosema-disease-of-honey-bees>

Spring expansion of the hive

In late August as the weather becomes milder, temperatures begin to exceed 18 degrees, pollen starts to become abundant and some nectar is available, a hive will show a significant increase in activity with many worker bees flying from, and returning laden, to the hive.

These factors combine to stimulate the queen to rapidly increase her rate of laying. Whilst a "full depth" frame has approx. 6500 cells, a queen will utilize only about 4500 of these for laying. (The remainder being used for pollen and honey storage for the brood food.). The queen can lay up to 1,500 / 2,000 eggs per day, thus may fill one frame in 2 - 3 days. In the 21 days it takes the worker larvae to mature and hatch, the queen may have used up all available cell space in an eight frame brood box.



It is critical to ensure over these months that the queen has adequate laying room or she may well decide to leave the hive with a swarm, seriously hindering the chances for a strong colony and honey collection season. Regular checking of the hive, at least every 12 days is essential to ensure there is adequate laying room for the queen, and to detect and remove any developing queen cells (which forewarn of swarming).

Comb replacement

The queen is much more stimulated to use fresh, clean frames of foundation. Therefore at this time of the year you should remove any old, dirty, or damaged frames from the brood box. These can be moved to the super for honey storage if required, or removed from the hive completely, cleaned and recycled.

Never separate existing frames of brood when adding new frames of foundation, as this makes it much more difficult for the nurse / house bees to maintain the warmth necessary for the larvae to develop. Always place the new frames of foundation to the outside of existing frames of brood.

When to add a super

Determining when to add a super to an existing single brood box is dependent on numerous factors. It should be done when six or more of the frames in the brood box are covered in bees and brood, or if when you lift the lid off a hive, the tops of approximately six or more frames are covered with bees.

If you do add a new super in spring, remove two or more of the outside frames from the brood box which contain brood and place into the centre of the super you are adding on. This encourages bees to go into the newly added top box. Replace the frames removed from the brood box with fresh foundation, or drawn comb, preferably from the same hive.

Reference: Australian Beekeeping Guide (RIRDC) Chp6



[Pre Spring Inspection video](#)

Pre-spring inspection



The video plays here in the reader. On paper: youtube.com/watch?v=3EgHwABJV_k

Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › [Resources](#) › [General Beekeeping](#) › [Spring hive development](#)

7 ways to find the queen

(And one way to get a result without finding the queen)

1. Go the centre

- Use minimal smoke, 1–2 puffs; take off the honey super and queen excluder.
- Space frames apart in the middle of the brood box.
- Scrape off burr comb on the side of the top bar.
- Carefully lift one frame out of the middle, taking care not to squash bees.
- Briefly scan over each side of this frame (like reading, taking about 1 minute). Place this frame in a spare box and continue with the next frame.

The queen can be on the wall of the hive body or on the bottom board. The centre of the brood nest, on a frame with eggs, is the most likely place to find her. Speed is an advantage because the old queen can hide well. Have an empty cage handy: if you find her on the first frame you might recycle her into a drone layer or an eggless hive later, if she has a good brood pattern or other good qualities. Once you've found her, shake bees off combs, check for disease, manipulate combs, scratch honey in corners to make egg-laying space, lift honey out to the top and replace with empty brood combs, introduce the cage and mark the front of the box with code – date – breeder (W123).

2. Go the outside

Otherwise known as the light barrier or Linton Briggs method.

- Place an empty box next to the hive.
- Use minimal smoke, 1–2 puffs; take off the honey super and queen excluder.
- With the sun over your left shoulder, remove the outside frame furthest from you, checking for the queen before placing it in the empty box.
- Remove the outside frame closest to you, again checking for the queen before placing it in the spare box. While it is not usual to find the queen on the outside frames (it can happen), removing these frames first creates a light barrier between the next frame and the hive wall. This confines the queen to the remaining frames.
- Before checking both sides of the next frame closest to you, glance down the face of the frame. According to Linton the queen often stands taller than the rest of the bees, and can be spotted up to 20% of the time on the face of the frame before it is removed. Repeat for the remaining frames until the queen is found.

3. Divide and conquer

- Put an empty box next to the hive.
- Take out half the frames and place them in the empty box. Put empty combs in each box.
- You will then have 4 frames of bees and brood, and 4 frames of pollen, honey and empty combs, in each box.

The next day one of the hives will have fanning bees at the front entrance (looking to attract the queen). The queen will be in the hive with no fanning bees at the entrance. She will now be easier to find because you know which box she is in and have only half the bees to search.

If you have 2 brood boxes, splitting them gives a similar result: half the bees with the queen in one box. The beauty of this method is that it only takes seconds to do.

4. Last resort

This one is a last resort and perfect for a recycled queen; it is suitable to use on a drone layer or a stinging hive.

- Move the whole hive 10–20 metres or more, preferably behind landmarks. This will cause the bees to drift back to their original position.
- Shake all bees off the combs onto the ground and place all frames into a new hive body elevated on a table. This ensures that the queen cannot re-enter the hive.
- Replace this brood box on the original spot to collect all the field bees.
- The queen will not be able to fly back.
- Check 7 days later for queen cells, knock them off and introduce a caged queen or cell.

5. Drift method

Suitable for pallets or apiaries with pairs or rows of hives.

- Move the hive to a new position a few metres away behind a landmark and turn the entrance 180 degrees.
- This will cause all the field bees to return to the hive next door. Usually this is OK under good conditions.
- The next day, or even a few hours later, you only have nurse bees and the queen on the brood frames; this gives you the edge in finding the old queen.

6. Divide and divide again

- Place an empty hive body next to the hive.
- Use method 1 first.
- Place four frames in each box with bees adhering. Pair up the frames. This should make the bees and the queen go between either of the two pairs of frames, where it is darker. After 10 minutes look for the queen, and remember to look on the wall and floor as well.

7. Strainer method

This one will demoralise the hive for a while, so should only be used if all else fails.

- Move the hive onto the next hive and put an empty body on the original place. On top of that place a super with the queen excluder screwed on the bottom (the strainer box).
- Shake all the bees into this strainer box, then return the brood frames from the original box, under the strainer, into the brood box on its original location.
- With a little smoke, force the bees through the strainer back into the brood box.

- You will only have drones and the queen left in the strainer. Use only cool smoke, so the bees don't panic and run up the walls of the strainer.

A slight variant is the Bob McDonald method. Move the hive and put down a new bottom board, put a queen excluder on the bottom, then place an empty box; shake all the bees in front of the original position, placing the brood frames into this box. The bees will enter, moving up through the queen excluder to look after the brood. When the hive has settled down, lift the brood box off the excluder and the queen will be trapped between the excluder and the bottom board. Place this bottom board and excluder in a hessian bag and remove.

8. Supersedure method (autumn only)

This method involves splitting the hive to introduce queen cells, possibly leaving the old queen.

- Shake all the bees off the brood combs into the old brood box and place the combs into a new super (not to be done under a heavy honey flow, because you risk drowning bees in their nectar).
- Fill the brood box with empty brood combs, replace the queen excluder, then place the box of brood on top. Nurse bees will move up through the queen excluder to feed the young. This can be done up to 7 days before.
- The old queen will occupy the empty combs and start laying there. After 6 hours or the next day the hive can be split with a division board or taken away to a new site. Either way you introduce the new queen into this brood box. Since the old queen has lost brood and bees she should be easier to find in the next few days.

By introducing a queen cell placed in a cell protector (or hair roller) it should survive and hatch the next day. When the virgin queen hatches she does not yet have queen pheromones, and the old queen is used to bees walking over her and grooming her, so the chance of the old queen being killed by the virgin queen's sting is very high. This method is widely used by commercial beekeepers overseas as it is very quick and should give 80% success, so I would split stronger hives to give over 100% laying young queens for the coming season. For this to work we need a good drone population and a breeding flow of nectar and pollen.

From the club's General Beekeeping documents: "7 Ways to Find the Queen" (PDF, 2 pages).

Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › [Resources](#) › [General Beekeeping](#) › [General Beekeeping](#)



National Bee Unit

Swarm Control When You Can't Find the Queen

This system can only be used if the colony is showing signs of swarming, i.e. there are queen cells with larvae in them. If queen cells are sealed it is too late as this indicates that the colony may have already swarmed.

1. Move the original brood box and floor to one side. A minimum of one metre. This is called the 'Parent Colony'; the younger bees remaining in this box;
2. Place a new brood box and floor filled with drawn comb or foundation on the original site. This is called the 'Artificial Swarm'; the flying bees will fill this box;
3. Examine the brood combs in the original brood box (parent colony) and remove a comb containing eggs and young brood. It must not have any queen cells on it;
4. Place this comb in the centre of the new box (artificial swarm);
5. Replace the vacant space in the original box (parent colony) with a new drawn comb or foundation;
6. Place any supers on the new brood box (artificial swarm) at the original site;
Leave for seven days then
7. Examine the new brood box (artificial swarm). If there are no queen cells then the queen is probably in this colony and eggs should be visible. If there are queen cells select one with a visible larva and destroy the rest;
8. If the queen is not present in the original box (parent colony) then emergency queen cells will be built. Knock down all queen cells bar one opened cell with a visible healthy larva. From this, a virgin queen will hatch and take over the colony. However, if the queen is present then due to the reduced number of bees they will break down any queen cells and eggs and young larvae will be seen;
9. Later in the season or in the following spring you will need to find the older queen to cull her. The colonies can then be re-united;

National Bee Unit

APHA, National Agri-Food Innovation Campus

Sand Hutton, York. YO41 1LZ

Telephone: 03003030094 email: nbuoffice@apha.gov.uk NBU Web site:

www.nationalbeeunit.com

January 2019

©Crown copyright. This sheet, excluding the logo, may be reproduced free of charge providing that it is reproduced accurately and not used in a misleading way. The material must be acknowledged



National Bee Unit

Shook swarm

August 2024

Shook swarms involve transferring adult bees onto fresh foundation in a clean hive and then destroying the old comb and any associated brood. Shook swarms are both an effective means of controlling pests and diseases and can be used to replace old combs. This fact sheet describes how to perform a shook swarm.

Why shook swarms are used

In the UK, shook swarms are used to treat colonies suffering from low levels of the brood disease, European foulbrood (EFB), replacing the use of antibiotics which are less effective¹. However, in the UK, EFB is a statutory notifiable disease and if EFB is suspected in a colony, beekeepers are legally obliged to inform the NBU. Shook swarms for the treatment of EFB are performed only by, or alongside, an Appointed Bee Inspector (ABI). **The protocol described below is for shook swarms where EFB is not present.**

It has been shown that a shook swarm is effective to reduce levels of *Varroa*², and can be used to treat colonies suffering from heavy chalkbrood infection³. It is also well worth performing a shook swarm if the colony has a lot of old and dirty brood frames.

When shook swarms can be carried out

The best time to carry out a shook swarm is between mid-March and mid-July, depending on the locality. The colony needs a substantial population of young bees at the right age to make wax; the majority of wax building bees are between 10 and 20 days old⁴. It is usually too cool for bees to effectively draw new comb before March and the loss of brood can be detrimental so early in the year. If a shook swarm is being used as a routine management tool, then it is best to avoid the period when eggs are laid that will develop into bees that will forage on the main honey flow. This will depend on the beekeeper's geographical region and experience of local main crops.

¹ Waite, R.J., Brown, M.A., Thompson, H.M. and Bew, M.H. (2003) Controlling European foulbrood with the shook swarm method and oxytetracycline in the UK. *Apidologie*, 34 (6). DOI: doi.org/10.1051/apido:2003052

² Mortensen, A.N., Jochym, M., Dearden, P.K. and Sainsbury, J.P. (2023) Old strategies as modern solutions, shook swarming reduced *Varroa destructor* infestation rates of managed honey bee, *Apis mellifera*, colonies. *Journal of Apicultural Research*, 1-7. DOI: doi.org/10.1080/00218839.2023.2243044

³ Betts, A. (1935) *The diseases of bees: their signs, causes and treatment*. Published by The Apis Club, UK.

⁴ Goodman, L. (2003) *Form and function in the honey bee*. Chapter 8: Glands. Jointly published by The International Bee Research Association and Northern Bee Books.

Preparation

Before getting started, you will need the following materials:

- A replacement brood chamber filled with clean frames of fresh foundation, floor and entrance block, crown board and roof
- A framed queen excluder
- A rapid feeder with capacity for at least 4.5 litres of feed
- Foam strips or other material for blocking entrances
- Strong plastic bags or paper bags large enough to place combs/frames and parcel or gaffer tape to seal them
- A queen cage (optional)

You will need to:

- Prepare 4 litres of strong sugar syrup by dissolving 1 kg of granulated white sugar per 500 ml of warm water
- Ensure all equipment is clean before use

Method for shook swarming

Shook swarming is not recommended for small colonies of less than 6 frames of bees. If shaking a small colony is necessary, then transfer them into a nucleus box.

Step 1. Move the hive that you intend to shake a short distance away from its original site



Figure 2: Ensure the queen excluder is beneath the brood box before shaking the bees in; this prevents the bees from absconding with the queen



Figure 1: Assemble a clean hive where the hive you intend to shook swarm was sited; move the original hive a short distance

Step 2. Place the clean floor on the original hive site. Put the queen excluder over the floor and then place the clean brood box containing frames of foundation on top of the queen excluder. Use an entrance block to support the queen excluder if it sags at the hive entrance on the original hive site

Step 3. Remove the centre four frames from the new brood box and put them to one side. The flying bees from the old hive will return to this clean brood chamber

Step 4. Open the hive you intend to shake, removing the supers and queen excluder (if present). Place them inside the upturned roof. Examine the old brood chamber and find the queen

Note: If you are not comfortable with finding the queen, then ensure you carefully shake the bees into the new colony and carefully check every piece of kit before reassembling it. When shaking frames, they should not be literally shaken but rather the bees are knocked off by a quick downward motion of the frame; being careful to avoid knocking the frame against the hive or other frames (see below)

Step 5. Once the queen has been found, place her between the frames of foundation in the new box, or in a queen cage so that she can be released into the new chamber when you have finished shaking the bees

Step 6. Shake all the bees from the old combs into the new chamber. This is done by holding a frame of bees about one third of the way into the gap left between the frames of foundation in the clean chamber. The frame is then moved quickly downward and suddenly stopped. Try to avoid knocking the comb against the chamber. The bees will fall off the comb into the new box and any remaining on the comb can be brushed off

Step 7. Shake and brush bees on the old equipment into the new chamber. If there are any supers, shake the bees from them into the brood box



Figure 3: Tightly grip the lugs of the frame and lower it gently into the space between the frames. Give the frame a sharp downward shake into the brood chamber

Step 8. When all the bees have been shaken into the new brood chamber, replace the frames of foundation that were removed to create space. Rest them into the gap and they will slowly sink down as the bees disperse upwards from the floor onto them. Take the queen out of the cage and place her into the brood box if she has been caged during the process.

Step 9. Put the crown board in place

Step 10. Feed the colony with a 'heavy' sugar syrup. Any supers can be placed above the feeder until the feeder is no longer needed; then it can be placed above the queen excluder as normal

Step 11. Inspect the colony after a week to check for brood. When brood is present, the queen is unlikely to abscond, remove the queen excluder and clean it appropriately. If there is no brood present, check again in another week. Only remove the queen excluder when you can see eggs and larvae

Step 12. Maintain feeding until all combs are drawn out. At this point, the end combs may need to be turned around or moved one frame into the chamber. This is because bees find it hard to cluster and create wax on frames adjacent to the side of the brood box



Figure 4: Once the bees have been shaken into the new brood box, the remaining frames of foundation can be carefully placed inside the box

Aftercare

After a colony has been shook swarmed, it will need to be checked regularly to ensure the colony recovers its strength. For more detailed information on how to care for a colony after a shook swarm, please see our fact sheet on ['Care of colonies after a shook swarm'](#).



Figure 5: After a shook swarm, the colony will benefit from having clean frames and freshly drawn-out wax. The colonies in these photos are 16 days post shook swarm

National Bee Unit

APHA, Room 11G03, York Biotech Campus,

Sand Hutton, York YO41 1LZ

Telephone: 0300 303 0094

email: nbu@apha.gov.uk

Web site: www.nationalbeeunit.com

August 2024

© Crown copyright 2024

You may re-use this information (excluding logos) free of charge in any format or medium, under the terms of the Open Government Licence v.3. To view this licence visit

www.nationalarchives.gov.uk/doc/open-government-licence/version/3/ or email

PSI@nationalarchives.gsi.gov.uk

Total Fire Ban Days: use of smokers

If you want to inspect your hives and use your smoker over summer, PLAN AHEAD!!!!

You are NOT permitted to light / use a smoker on a total fire ban day.

Anyone who breaches a Total Fire Ban declaration is subject to heavy legal penalties including large fines and even imprisonment.

<https://www.cfa.vic.gov.au/warnings-restrictions/total-fire-bans-fire-danger-ratings/about-total-fire-bans>

Summer - If you want to inspect your hives and use your smoker over summer, PLAN AHEAD!!!!



CFA declares the Fire Danger Period for each municipality (shire or council) at different times in the lead up to the fire season. It depends on the amount of rain, grassland curing rate and other local conditions.

The Fire Danger Period may be declared as early as October in some municipalities, and typically remains in place until the fire danger lessens, which could be as late as May.

The Fire Danger Period is NOT the same thing as a Total Fire Ban.

Your local council or shire may have their own local laws about lighting fires. [Contact your local council](#) to find out what municipal local laws may be in place. Local laws may be in place all year.

You are NOT permitted to light/use a smoker on a total fire ban day.

Anyone who breaches a Total Fire Ban declaration is subject to heavy legal penalties including large fines and even imprisonment.

Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › Resources › General Beekeeping › Total Fire Ban Days - Use of Smokers

What to do when you are Queenless

This article was originally published in May 2013*, but this issue comes up like clockwork every year...

(*If any one knows who the original author of this article is, please contact us so we can publish the appropriate credit)

You think your hive is queenless - you can't spot the queen, and you don't see any eggs. What now?

First, don't panic - unless you have been pretty negligent in your inspections you have plenty of time to address this issue without any real negative effects on the hive. There is at least a 3-1 chance that your hive is not actually queenless, and even if it is you can get it queenright all by yourself if you have at least one other hive.

Give the "queenless" hive a frame from another hive that has at least a few eggs or very young brood on it - aren't you glad you have more than one hive? Mark that frame (a child's crayon works great for this) and make a note of what you have done and the date. If the hive really is queenless, then it will start queen cells on that frame of brood right away, and they will be easy for even a novice to spot within 3-5 days. The only real exception is that a hive which has been actually queenless for more than about 30 days may be so weak or have so many laying workers already that they won't start queen cells. By that time though the hive will not only be queenless but will be completely broodless and probably very weak - many of the foragers will abandon a queenless hive. But if you do inspections at least every other week to confirm that your hives are queenright that will never happen.

If they don't try to start queen cells on a fresh frame of brood (during Spring through Fall) then they already have a queen - it might be a virgin queen or a recently mated queen that is not yet laying. If they do already have a queen (of any kind) they WILL NOT accept a new queen - no matter how much you pay for it.

Just about the only ways to be sure that a hive is really and truly queenless is to do the frame-of-brood thing or to actually remove the queen yourself. Looking for the queen doesn't do it - even an experienced queen spotter can fail when it really matters.

Also, giving an actually queenless hive a frame of open brood will help to prevent it from developing a laying worker, and will help to keep the population demographics healthier.

Any hive will benefit from a donated frame of brood.

There are a few reasons why a hive might appear to be queenless when it really isn't. Brood production is cyclical - a hive can be full of brood one day and have very little only a few days later. This usually involves an influx of nectar that leaves the queen with few places to lay eggs - at which point the hive may appear to be without a queen. Depending on the exact circumstances this may or may not result in the hive issuing a swarm. A hive which has swarmed may be queenless for a few days, and may be broodless for up to 3 weeks or so while it makes a new queen - also appearing to be queenless. Or you could accidentally squish the queen during an inspection. None of these circumstances make a hive hopelessly queenless - they all will make a new queen most of the time without beekeeper intervention. Sometimes (10-30% of the time) something goes wrong though - If a virgin queen fails to return from mating for any reason the hive will then be hopelessly queenless - lacking the resources to make a new queen without help from a beekeeper. While a

queenless hive will pretty much always try to make a new queen it takes about 24 days more or less for that new queen to develop, get mated, and start laying eggs. For many people - myself and my 50 yr old eyes included - it will be another week before there is brood which is easy to spot. So about a month between becoming queenless and easily spotting brood. During that time all of the eggs that the previous queen laid will emerge leaving the hive completely broodless after 24 days - all of the worker brood emerges in 21 days leaving only capped Drone brood. This can make you think that you have a laying worker or drone laying queen.

Whenever in doubt - give any possibly queenless hive a frame of open brood. Buying a queen without running this test first is going to be a waste of money about 3 out of 4 times.

Timeline of Queenlessness

No brood of any kind, population weak, laying workers, SHB, robbers, or wax worms taking over - queenless too long to save in my opinion. Shake it out - it's a lost cause.

No brood of any kind but population strong - hive has been queenless for over three weeks - at least 24 days. If the population is still strong and you can see where they have cleaned out comb for a queen to lay eggs, there is probably a queen that either hasn't started laying yet or has laid eggs that you are not spotting. Giving it a frame of brood is good luck anyway.

Capped Drone brood only - hive has been queenless for just about 3 weeks.

Lots of capped worker brood, but no open brood at all - queenless for about 1 1/2 - 2 1/2 weeks.

Open larva but no eggs or young brood - Queenless 6-8 days. You should find capped queen cells in a hive like this.

If I made a hive queenless, then I usually try to leave it alone for about 3-4 weeks if I can remember exactly when I did it. I need to keep better records I know. If I find one that looks like it has been queenless for only a couple of weeks or less I look for cells and then leave it alone for a couple of weeks. I always give a hive which has been queenless for over 3 weeks (little if any worker brood) a frame of young brood from another hive to see if it builds new cells or so as to confirm if it is still queenless - and to ward off laying worker.

Remember - it takes a hive about 12 days to raise a queen, but it takes that queen another week to harden up and get mated, and then another week to start laying. Then it might be another week before you can spot any brood. About a whole month from start to finish to produce an easy to find laying queen.

But it only takes a few minutes to give a hive a frame of brood - and avoid disaster.

And by the way - you will not hurt the donating hive by stealing one frame of brood from it - even if you do it every week for a while. Fear of harming a hive by taking brood from it is very common but almost completely baseless. Don't let this imaginary harm cause you to needlessly lose another hive. If it bothers you then place back the donation once you get the other hive queenright and healthy again. But most likely you will see then that there is no need to do so because stealing a frame of brood here and there does little to set back a hive.

Shake it out!

Caution - if a hive has really been queenless for a whole month it is only a few days from a total meltdown which often results in the loss (due to wax moths and hive beetles) of all hive resources - including comb. There is a point (little or no brood) when the best thing to do is to shake out the hive (carry it 15 yards away and shake all of the bees off of every frame onto the ground - then remove all of the woodenware from the original spot where it sat) and freeze the frames to kill any hive beetle or wax moth eggs which may be present. If you have to shake out a hive it is not at all a loss, but rather more like a reboot - no bees (except laying workers - all others will beg into other hives) are killed, and no comb, honey, or brood is lost. After you freeze the frames you can redistribute them and start a new nuc in the same spot using resources from your other hives, and many of the old foragers will even return to it.

Do your inspections!

Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › Resources › General Beekeeping › [What to do when you are Queenless](#)

Winter pack down guide

[Winter welfare for your bees video](#)

The time to perform a winter pack down is around April 25th

- 1 Have all equipment you require including extra boxes including storage boxes. It is important to have a plan of how and what you are preparing to do.
- 2 Set up your gear around the hive so once you start you are not trying to look for anything. It is a good idea to have three areas to put the frames you remove from the hive: one for brood frames, one for honey frames and the other for empty frames.
 
- 3 Smoke the bees. Wait a minute for it to take effect.
- 4 Remove the lid. It is a good idea to turn the lid upside down and put a spare super on it. This is where you put your brood frames. The reason for this: if you accidentally have the queen on one of your frames and she drops off, she will land in the lid and not on the ground.
 
- 5 Look into the top of the box and count the frames of bees. The frame should be fully covered on both sides with bees to be a 'full frame'. Otherwise, gauge from two or more frames with partial bee numbers to 'estimate' the number of full frames you have. This will depend on how far you can pack the bees down.
 
- 6 Start removing the frames as you would if you were checking the hive, except separate the frames - brood in one box, honey frames in another box and the empty frames in another box. Keep trying to find the queen on each frame until located.
 

7

Remove the empty super and view the next super (or brood box if no other supers) - again look into the top of the box and count the frames of bees. This will depend on how far you can pack the bees down.



8

Start removing the frames as you did in Step 6 - separating the frames - brood in one box, honey frames in another box and the empty frames in another box.



9

Continue steps 5 to 8 until you have inspected all the supers and brood box.

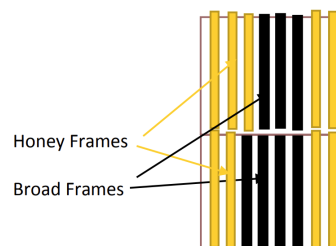
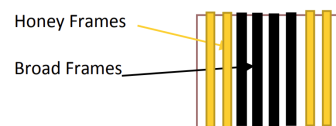


10

Now count up your frames to estimate the total number of frames of bees. This is a guide to the number of honey frames required. Try and have 1 brood frame to 1 capped honey frame.

11

Start replacing the frames. If you are packing down to one super - positions 1 & 8 are usually for honey frames, then from position 7 to 2. Try and have the honey frames on the outside and the brood frames in the middle. If you are packing down to 2 supers - the main preference is to have the majority of the brood frames in the bottom super. Say 5 in the bottom and 3 in the second super, always in the centre. Also ensure there is space for the queen to lay. Do not put frames with a lot of nectar as this may ferment over the cooler months and cause nosema or other issues. If you are packing down to 2 supers it is a good idea to remove the queen excluder to allow the queen to roam between both boxes.



12

Once you have all frames in place, put your hive mat and lid in place. Secure your hive with an emlock or other method. At this point lift the back of the hive and feel the weight. You will perform this over winter to check to see if the bees have food. As food is consumed the weight will lessen and if too light, you may need to feed the bees. Please note, if you don't have enough honey frames as above, you will have to monitor the weight of the hive and feed them during winter.



13

Remove all surplus frames and boxes. If you have honey frames, you can extract or store for the bees in early spring; remove all nectar from the frames. It is best to freeze all these frames for at least 3 days to kill off any wax moth larvae. Store in clear, sealed containers.



[Winter pack down video](#)

Winter pack down video



The video plays here in the reader. On paper: youtube.com/watch?v=NbywGTsFYWs

Source: the Geelong Beekeepers Club website, geelongbeekeepersclub.org.au › [Resources](#) › [General Beekeeping](#) › [Winter packdown](#)

Flow Frame Cleaning

Jurg Schutz

How to clean Flow frames, demonstrated by club member Jurg Schutz.



The video plays here in the reader. On paper: geelongbeekeepersclub.org.au/edit/movie/Flow-Frame_Cleaning.mp4

Source: Geelong Beekeepers Club website

Varroa Life Cycle

The life cycle of the varroa mite inside the brood cell. Courtesy of the Queensland BeeAware website.



The video plays here in the reader. On paper: [geelongbeekeepersclub.org.au/edit/movie/varroa life cycle.mp4](http://geelongbeekeepersclub.org.au/edit/movie/varroa%20life%20cycle.mp4)

Source: Geelong Beekeepers Club website

Winter welfare for your bees

Keeping colonies well through the cold months.



The video plays here in the reader. On paper: [geelongbeekeepersclub.org.au/edit/movie/winter welfare for you bees.mp4](http://geelongbeekeepersclub.org.au/edit/movie/winter%20welfare%20for%20you%20bees.mp4)

Source: Geelong Beekeepers Club website

What is honey bee biosecurity?

BeeAware: Bee Biosecurity Video Series, video 1



The video plays here in the reader. On paper: youtube.com/watch?v=Ylg4qjJ8cH4

Source: BeeAware: Bee Biosecurity Video Series, video 1

National Bee Biosecurity Program and Code of Practice

BeeAware: Bee Biosecurity Video Series, video 2



The video plays here in the reader. On paper: youtube.com/watch?v=TZNKg6Mx40c

Source: BeeAware: Bee Biosecurity Video Series, video 2

National Bee Pest Surveillance

BeeAware: Bee Biosecurity Video Series, video 3



The video plays here in the reader. On paper: youtube.com/watch?v=eE7wKn0ul74

Source: BeeAware: Bee Biosecurity Video Series, video 3

Alcohol wash test for varroa mite

Tocal College / NSW DPI



The video plays here in the reader. On paper: [geelongbeekeepersclub.org.au/edit/movie/alcohol wash - tocal nsw dpi.mp4](http://geelongbeekeepersclub.org.au/edit/movie/alcohol%20wash%20-%20tocal%20nsw%20dpi.mp4)

Source: Tocal College / NSW DPI

Sugar shake test for mites (Ag NSW)

Tocal College / NSW DPI



The video plays here in the reader. On paper: [geelongbeekeepersclub.org.au/edit/movie/sugar shake - tocal nsw dpi.mp4](http://geelongbeekeepersclub.org.au/edit/movie/sugar%20shake%20-%20tocal%20nsw%20dpi.mp4)

Source: Tocal College / NSW DPI

Sugar shake test for mites (Ag Vic)

Agriculture Victoria



The video plays here in the reader. On paper: youtube.com/watch?v=i0SgirFgFck

Source: Agriculture Victoria

Drone uncapping test for varroa mite

Agriculture Victoria



The video plays here in the reader. On paper: youtube.com/watch?v=403G94pmewQ

Source: Agriculture Victoria

Varroa: cultural control

Non-chemical varroa control techniques



The video plays here in the reader. On paper: [geelongbeekeepersclub.org.au/edit/movie/cultural control of varroa.mp4](http://geelongbeekeepersclub.org.au/edit/movie/cultural%20control%20of%20varroa.mp4)

Source: Non-chemical varroa control techniques

Small hive beetle

Agriculture Victoria



The video plays here in the reader. On paper: youtube.com/watch?v=p82pNZXwVv8

Source: Agriculture Victoria

Small hive beetle trap / attractant

Agriculture Victoria



The video plays here in the reader. On paper: youtube.com/watch?v=YHUmK5SlzXU

Source: Agriculture Victoria

Small hive beetle control devices

Agriculture Victoria



The video plays here in the reader. On paper: youtube.com/watch?v=M8_8Svz9F5s

Source: Agriculture Victoria



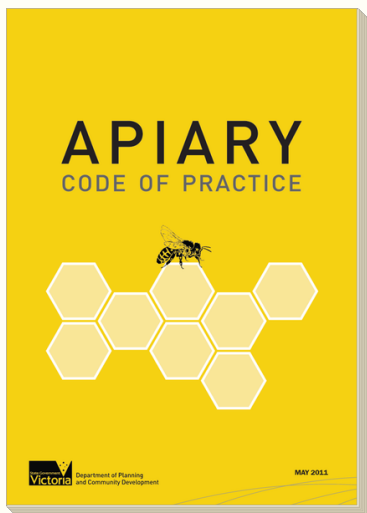
Plan Bee: Honey Bee Breeding Manual

37 pages • 5.3 MB

AgriFutures Australia's manual from the national Plan Bee genetic improvement program: selecting, recording and breeding better bees. September 2025 edition.

Open the document ›

Opens in a viewer over the reader; on paper, the file is at [geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/AgriFutures Plan Bee Breeding Manual 2025-09.pdf](https://geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/AgriFutures%20Plan%20Bee%20Breeding%20Manual%202025-09.pdf)



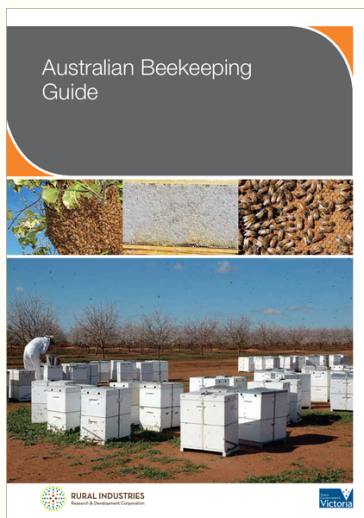
Apiary Code of Practice

20 pages · 1.9 MB

The Victorian Apiary Code of Practice: the rules for keeping bees on urban and rural land, from hive numbers and placement to water, swarm control and nuisance.

Open the document ›

Opens in a viewer over the reader; on paper, the file is at [geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Apiary Code of Practice.pdf](http://geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Apiary%20Code%20of%20Practice.pdf)



Australian Beekeeping Guide

PDF • 6.3 MB

The RIRDC/AgriFutures guide for beginning and established beekeepers: equipment, handling, seasonal management, pests and diseases, and honey.

Open the document ›

Opens in a viewer over the reader; on paper, the file is at [geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Australian Beekeeping Guide.pdf](http://geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Australian%20Beekeeping%20Guide.pdf)



Australian Honey Bee Industry Biosecurity Code of Practice

21 pages • 4.1 MB

The national code every registered beekeeper is expected to follow: inspections, record keeping, pest surveillance and reporting. October 2025 revision.

Open the document ›

Opens in a viewer over the reader; on paper, the file is at [geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Australian Honey Bee Industry Biosecurity Code of Practice 2025-10.pdf](https://geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Australian%20Honey%20Bee%20Industry%20Biosecurity%20Code%20of%20Practice%202025-10.pdf)



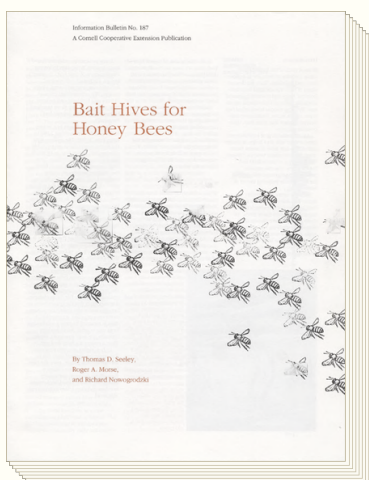
Australian Honey Bee Industry Biosecurity Manual

88 pages · 15.3 MB

The companion manual to the Code: how to recognise and respond to the pests and diseases of honey bees. October 2025 revision.

Open the document ›

Opens in a viewer over the reader; on paper, the file is at [geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Australian Honey Bee Industry Biosecurity Manual 2025-10.pdf](https://geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Australian%20Honey%20Bee%20Industry%20Biosecurity%20Manual%202025-10.pdf)



Bait Hives for Honey Bees

by Thomas D. Seeley · 8 pages · 824 KB

Tom Seeley's classic Cornell extension bulletin on what swarms look for in a home, and how to build a bait hive that catches them.

Open the document ›

Opens in a viewer over the reader; on paper, the file is at [geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Bait Hives for Honey Bees - Tom Seeley.pdf](http://geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Bait%20Hives%20for%20Honey%20Bees%20-%20Tom%20Seeley.pdf)



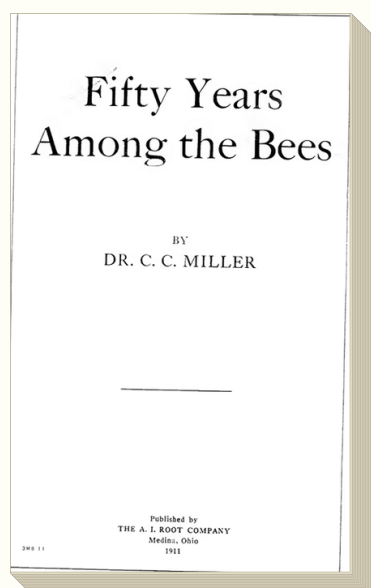
Bee Friendly: a planting guide for European honeybees and Australian native pollinators

by Mark Leech • 330 pages • 8.1 MB

The RIRDC planting guide by Mark Leech: which trees, shrubs and garden plants feed bees across Australia's regions, and when they flower.

Open the document ›

Opens in a viewer over the reader; on paper, the file is at [geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Bee Friendly Planting Guide.pdf](http://geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Bee%20Friendly%20Planting%20Guide.pdf)



Fifty Years Among the Bees

by C. C. Miller · 364 pages · 19.5 MB

Dr C. C. Miller's 1911 account of a lifetime of comb-honey beekeeping in Illinois — still read for its good sense on bees and on beekeepers.

Open the document ›

Opens in a viewer over the reader; on paper, the file is at [geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Fifty years among the bees - C C Millar.pdf](https://geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Fifty_years_among_the_bees_-_C_C_Millar.pdf)



Queen cells in my hive

25 pages · 735 KB

How to tell swarm cells from supersedure and emergency cells, and what each one is telling you to do.

Open the document ›

Opens in a viewer over the reader; on paper, the file is at [geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Queen cells in my hive.pdf](http://geelongbeekeepersclub.org.au/edit/general_beekeeping_documents/Queen%20cells%20in%20my%20hive.pdf)

Elsewhere on the website

The club keeps its varroa material together on the website: treatments and the current advice for the Geelong region, monitoring methods, the GBC Varroa Watch, and the latest from Agriculture Victoria.

Varroa Management Hub ›

geelongbeekeepersclub.org.au › Resources › Varroa management

Bee biology in the newsletter ›

Bruce Ward's "This month's bit of bee biology" series and everything else tagged Bee biology in the club newsletter reader.

These pages were compiled from [the General Beekeeping page](#) on geelongbeekeepersclub.org.au, where the downloadable documents live.